1) $(x^3 - 7x^2 - 6x + 5) \cdot (-7x^3 - 2x^2 - 5x + 5) =$

$= -7x^6 + 49x^5 + 42x^4 - 35x^3 - 2x^5 + 14x^4 + 12x^3 - 10x^2 -$

$- 5x^4 + 35x^3 + 20x^2 - 25x + 5x^3 - 35x^2 - 30x + 25 =$

$= -7x^6 + 47x^5 + 51x^4 + 17x^3 - 15x^2 - 55x + 25$

2) $x^3 - 4x^2 + 3x = x(x^2 - 4x + 3) = x(x-3)(x-1)$

$x^3 - 4x^2 + 3x \;|\; x^2 - x = (x-1)x$

$x^3 - x^2 \;\;\; | \;\; x - 3$

$-3x^2 + 3x$

$-3x^2 + 3$

0

3) $\frac{-x^3-5x^2+x-10}{-4x^2+9x-4}=$

$$\begin{array}{r|l}
-x^3-5x^2+x-10 & -4x^2+9x-4 \\
\hline
-x^3+\frac{9}{4}x^2-x & \frac{1}{4}x+\frac{29}{16} \\
\hline
-\frac{29}{4}x^2+2x-10 & \\
-\frac{29}{4}x^2+\frac{261}{16}x-\frac{29}{4} & \\
\hline
-\frac{229}{16}x-\frac{11}{4} &
\end{array}$$

4) $x^4 - 2x^3 - 4x^2 + 2x + 3 \;\big|\; x^2 - 1$

$x^2 - 2x - 3 = (x-3)(x+1)$

$x^4 - x^2$

$-2x^3 - 3x^2$

$-2x^3 + 2x$

$-3x^2 + 3$

$-3x^2 + 3$

0

5) $x^2 + 12x + 61$

$$x_{\pm} = -\frac{12 \pm \sqrt{144 - 244}}{2} = -\frac{12 \pm 10i}{2} = -6 \pm 5i$$

6) $\begin{array}{l|l} x^5 - 7x^3 + 2x^2 + 12x - 8 & x^3 - 4x^2 + 5x - 2 \\ \underline{x^5 - 4x^4 + 5x^3 - 2x^2} & x^2 + 4x + 4 \\ 4x^4 - 12x^3 + 4x^2 + 12x & \\ \underline{4x^4 - 16x^3 + 20x^2 - 8x} & \\ 4x^3 - 16x^2 + 20x - 8 & \\ \underline{4x^3 - 16x^2 + 20x - 8} & \\ 0 & \end{array}$

7) $x^5+2x^4-16x^3-14x^2+63x-36=f(x)$

$f(-3)=\underbrace{-243+2\cdot 81}_{-81}+\underbrace{16\cdot 27}_{432}-\underbrace{14\cdot 9}_{126}-\underbrace{63\cdot 3}_{189}-36=$

315

$f'(x)=5x^4+8x^3-48x^2-28x+63$ (1)

$f'(-3)=\underbrace{5\cdot 81}_{405}-\underbrace{8\cdot 27}_{216}-\underbrace{48\cdot 9}_{432}+\underbrace{28\cdot 3}_{84}+63\neq 0$

8) $y=x^2(x+4)(x+1)=x^2(x^2+5x+4)=x^4+5x^3+4x^2$